Exact Log-Concavity Thresholds for Partitions into Fourth, Fifth and Sixth Powers

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Abstract

For $k \in \{4, 5, 6\}$, let $p_k(n)$ count unrestricted unordered partitions of n into positive perfect kth powers. We prove, by a computer-assisted argument, that the exact permanent log-concavity thresholds are 637855, 2507861 and 35577569, respectively. In each case the inequality is strict at every integer from the threshold onward, and an exact negative determinant at the preceding integer proves minimality. A unified argument combines explicit logarithmic curvature of a Wright-function model with a Mellin transformation and certified bounds on three contour regions. It proves strict infinite tails beginning at 2000000, 12000000 and 180000000. Exact finite coefficient reconstructions cover the joins and classify all equality cases. The finite integer arithmetic and the rational contour certificates have separately implemented checks; the scope and shared components of these checks are specified.

1 Definitions, target and prior work

Set

$$F_k(q) = \prod_{j \geq 1} (1 - q^{j^k})^{-1} = \sum_{n \geq 0} p_k(n) q^n, \qquad p_k(0) = 1,$$

and, for integers $n \geq 1$,

$$D_k(n) = p_k(n)^2 - p_k(n-1)p_k(n+1), \qquad T_k = \min\{t \geq 1 : D_k(n) \geq 0 \text{ for all integers } n \geq t\}.$$

All parts are positive, order is ignored, and repetitions are allowed. For example $p_k(2^k) = 2$. The strict threshold, with $D_k(n) > 0$, is a separate quantity, denoted T_k^{str}. Our result determines both.

Theorem 1.1. *For the unrestricted power-partition functions defined above,*

$$(T_4, T_5, T_6) = (637855, 2507861, 35577569), \qquad T_k^{\mathrm{str}} = T_k \quad (k = 4, 5, 6).$$

The respective numbers of positive integers with $D_k(n) = 0$ are 525*,* 2223 *and* 11411*; their largest elements are* 1118*,* 4098 *and* 25767*. The complete equality sets are specified by the exact finite certificate and verification procedure in Section 2.*

Roy's Conjecture 3.3 and Table 3 [1, 2] use the convention $n > N_k$, with $N_4 = 637854$, $N_5 = 2507860$ and $N_6 = 35577568$. Thus the candidates under our inclusive convention are

$$(T_4, T_5, T_6) = (637855, 2507861, 35577569). \tag{1}$$

The journal version published on 29 September 2026 continues to label these values as conjectural. Its earlier arXiv version 4 is dated 10 May 2025. O'Sullivan's Theorem 1.3 [3] already proves, for every fixed positive integer k and all sufficiently large n,

$$p_k(n)^2 > p_k(n-1)p_k(n+1)(1+n^{-2}).$$

Consequently T_k exists. That theorem does not give the explicit numerical cutoff required here. Corollary 5.7 of the same paper gives, for fixed integer d, with $A_k = \Gamma(1+1/k)\zeta(1+1/k)$, $a_k = (A_k/k)^{k/(k+1)}$ and $h_k = \zeta(-k)/2$,

$$\frac{p_k(n+d)p_k(n-d)}{p_k(n)^2} = 1 - \frac{ka_kd^2}{(k+1)(n+h_k)^{2-1/(k+1)}} + \frac{(3/2-1/(k+1))d^2}{(n+h_k)^2} + O_{k,d}(n^{-2-1/(k+1)}).$$

Here $h_4 = h_6 = 0$ and $h_5 = -1/504$. An unspecified constant in this remainder cannot join a finite calculation to an infinite theorem.

A source-specific audit on 24 October 2026 checked the latest versions of these papers and related threshold searches. No complete resolution of the three exact thresholds was located. This bounded search is not a guarantee of priority. No result from the author's companion square- and cube-partition studies is used as an axiom.

2 Exact finite results

Proposition 2.1. *The following statements hold on the indicated finite ranges. At every integer from the candidate threshold in (1) through B_k, the determinant $D_k(n)$ is strictly positive.*

k	B_k	*Negative count*	*Last negative*	*Zero count*	*Last zero*
4	*2000000*	*264873*	*637854*	*525*	*1118*
5	*12000000*	*974455*	*2507860*	*2223*	*4098*
6	*180000000*	*14548865*	*35577568*	*11411*	*25767*

In particular,

$$D_4(637854) = -592506620242280764650822263749 9,$$
$$D_5(2507860) = -154078232969963080665921 68,$$
$$D_6(35577568) = -621652977150045976221990 3616.$$

These are failure witnesses, and hence prove $T_4 \geq 637855$, $T_5 \geq 2507861$ and $T_6 \geq 35577569$.

2.1 Counting and storage bounds

For a coefficient limit $L = B_k + 1$, parts larger than L have no effect. If $C_j(n)$ counts partitions using $1^k, \ldots, j^k$, separating those partitions that contain a j^k from those that do not proves

$$C_j(n) = C_{j-1}(n) + C_j(n-j^k), \qquad C_0(0) = 1,$$

with $C_0(n) = 0$ for positive n and $C_j(n) = 0$ for negative n. Thus the ascending update $c(n) \leftarrow c(n) + c(n-j^k)$ computes the coefficients exactly. Each pass adds one allowed part, and induction over the passes establishes the invariant. The order of the allowed parts may be reversed, because it leaves the finite generating product unchanged.

An elementary rational bound proves that unsigned 128-bit storage suffices. Let $J = \lfloor L^{1/k} \rfloor$ and $1 \le s \le J$. The multiplicities of parts j^k for $j > s$ have at most $1 + \lfloor L/j^k \rfloor$ possibilities each. For the remaining multiplicities of $2^k, \dots, s^k$, attach disjoint half-open unit boxes to all nonnegative lattice points whose weighted sum is at most L. These boxes lie in the positive simplex of weighted size $L + \sum_{j=2}^{s} j^k$. Its volume bounds the number of lattice points. The multiplicity of 1 is fixed by all the others. Hence, for every $0 \le n \le L$,

$$p_k(n) \le \frac{(L + \sum_{j=2}^{s} j^k)^{s-1}}{(s-1)! \prod_{j=2}^{s} j^k} \prod_{j=s+1}^{J} \left(1 + \left\lfloor \frac{L}{j^k} \right\rfloor\right). \tag{2}$$

The same bound applies at intermediate passes, since restricting the allowed parts cannot increase a coefficient. Minimizing the displayed rational expression over $1 \le s \le J$ for $k = 5, 6$ gives integer upper bounds

$$23879217093903059394487995569892 62, \qquad 32525364573171940479329587296970884 57,$$

of bit lengths 111 and 122.

For the longer fourth-power range we use a sharper elementary bound. Write $F_k(e^{-t}) = \prod_{j \ge 1} (1 - e^{-tj^k})^{-1}$, $x = t^{-1/k}$, and $A_k = \Gamma(1 + 1/k)\zeta(1 + 1/k)$. The function $f(u) = -\log(1 - e^{-tu^k})$ is convex for $u > 0$: putting $v = tu^k$ gives

$$f''(u) = \frac{ktu^{k-2}}{e^v - 1} \left(\frac{kv}{1 - e^{-v}} - (k-1) \right) > 0,$$

because $1 - e^{-v} < v$. Midpoint convexity on each interval $[j - 1/2, j + 1/2]$ and the positive exponential series give

$$\log F_k(e^{-t}) \le \int_{1/2}^{\infty} f(u)\,\mathrm{d}u = A_k x - \int_{0}^{1/2} f(u)\,\mathrm{d}u \le A_k x + \tfrac{1}{2}\log t - \tfrac{k}{2}(1 + \log 2).$$

For the last step use $1 - e^{-tu^k} \le tu^k$. Adding ones injects partitions of n into those of $n + r$, so $F_k(e^{-t}) \ge p_k(n) e^{-nt}/(1 - e^{-t})$. Consequently

$$p_k(n) \le t^{3/2} \exp(nt + A_k x). \tag{3}$$

Logarithmic convexity of Γ between 1 and 2 gives $\Gamma(5/4) \le 1$. The integral test and elementary fourth-root comparisons give

$$\zeta(5/4) < 1 + 2^{-5/4} + \tfrac{13}{3} 3^{-1/4} < 1 + \tfrac{17}{40} + \tfrac{13}{3}\tfrac{19}{25} < \tfrac{19}{4}.$$

Thus, with $L = 2000001$ and $x = 18$, formula (3) is bounded by $18^{-6} \exp(L/18^4 + 171/2)$, whose upward rational enclosure has bit length 126. To compute that enclosure without a transcendental library, for $y \ge 0$ put $M = \lceil y \rceil + 100$ and use

$$e^y \le \sum_{r=0}^{M} \frac{y^r}{r!} + \frac{y^{M+1}}{(M+1)!} \frac{1}{1 - y/(M+2)}.$$

The bound follows by comparing all successive omitted-term ratios with $y/(M+2) < 1$; substituting the displayed rational y proves the bit claim by integer comparison with 2^{126}. Thus unsigned 128-bit storage suffices in every case. In addition to these a priori proofs, every computed addition checks for an overflow. Products of two unsigned 128-bit integers fit in 256 bits, so comparison of the two products in $D_k(n)$ is exact.

2.2 Independent checks and finite scope

The generator uses two 64-bit limbs. The checking traversal uses four 32-bit limbs, reverses the part order, regenerates every coefficient, and evaluates every determinant using arbitrary-precision integers. It checks each claimed sign and the displayed boundary values. The disk-backed sixth-power implementations share portions of the original block input/output and control machinery; their limb arithmetic and part order differ, while both disk scans evaluate determinants with arbitrary-precision integers. The final one-file checking reconstruction also has separately written I/O. They are not described as wholly independent software implementations. The sign at every index is retained in a compressed signed-byte stream; complete equality lists are also stored in the compact certificates.

More precisely, for each k the certificate contains a byte sequence $s_1, \dots, s_{B_k}$ with values $-1, 0, 1$ and an increasing list Z_k. The checker reconstructs $c_0, \dots, c_{B_k+1}$ by the proved finite product recurrence, verifies

$$s_n = \operatorname{sign}(c_n^2 - c_{n-1}c_{n+1}) \quad (1 \le n \le B_k), \qquad Z_k = \{n : s_n = 0\},$$

and verifies the counts, largest negative index and boundary coefficients. This specifies the essential certificate verification mathematically. The zero lists are exact enumerations, rather than samples; the analytic theorem below will show that they are the global equality sets. As simple checks on the convention, $D_k(n) = 0$ for $1 \le n \le 2^k - 2$, $D_k(2^k - 1) = -1$ and $D_k(2^k) = 2$.

There is a further, different-algorithm check. Formal logarithmic differentiation gives

$$\frac{qF_k'(q)}{F_k(q)} = \sum_{r \ge 1} b_k(r) q^r, \qquad b_k(r) = \sum_{j^k \mid r} j^k,$$

and therefore

$$np_k(n) = \sum_{r=1}^{n} b_k(r) p_k(n-r), \qquad p_k(0) = 1.$$

Each coefficient of these formal products is a finite sum. The recurrence uniquely determines the sequence by induction. A separate arbitrary-precision implementation verifies it through $n = 8192$, checking that each division has zero remainder. This independent recurrence check concerns that prefix; it is not described as a different-algorithm check of the entire finite range.

The proposition is a computer-assisted finite result, using the proved recurrences and the executed exact checks. The infinite-tail theorem is needed to exclude further negative or zero determinants beyond B_k.

3 A uniform curvature theorem for the smooth model

Fix $k \in \{4, 5, 6\}$, $A > 0$ and $m > 0$, and put

$$C = (2\pi)^{-k/2}, \quad X = (A/k)^{k/(k+1)} m^{1/(k+1)}, \quad z = Am^{1/k} = kX^{1+1/k}.$$

Define the entire Wright series and its coefficient model by

$$W(z) = \sum_{r \ge 0} \frac{z^r}{r!\Gamma(r/k - 1/2)}, \quad H(m) = Cm^{-3/2} W(z), \quad M(m) = Cm^{-3/2} X e^{(k+1)X} \sqrt{\frac{k}{2\pi(k+1)}}.$$

The reciprocal gamma function is its entire continuation, so terms at gamma poles vanish. Stirling's bounds show local uniform convergence of the series and its derivatives. The following theorem concerns this model; identifying and bounding its difference from $p_k(n)$ is a separate step.

Theorem 3.1. *For $X \geq 10$,*

$$H(m) > \frac{1}{5}M(m) > 0, \qquad -\frac{\mathrm{d}^2}{\mathrm{d}m^2}\log H(m) \geq \frac{((k-1)/k)X - 3/2}{m^2} > 0.$$

The positive part $H_+ = Cm^{-3/2}W_+$, defined below, also satisfies

$$H_+(m) > M(m)/2 \qquad (X \geq 15). \tag{4}$$

3.1 A lower bound

Write $W = W_+ - S$, where W_+ is the sum over $r > k/2$ and S is the absolute value of the remaining, nonpositive terms. For real $L = r/k > 1$, log-convexity of gamma gives

$$\Gamma(L - 1/2) \leq \sqrt{\Gamma(L-1)\Gamma(L)} = \frac{\Gamma(L+1)}{L\sqrt{L-1}}.$$

The upper Stirling inequality

$$\Gamma(v+1) \leq \sqrt{2\pi v}(v/e)^v e^{1/(12v)} \quad (v > 0)$$

then bounds an individual positive summand below by

$$W_r \geq \frac{\sqrt{L-1}}{2\pi\sqrt{k}} \exp\left((k+1)L(1 + \log(X/L)) - \frac{k+1}{12kL}\right). \tag{5}$$

The cited real-argument Stirling inequality follows, for example, from Binet's integral remainder: its positive remainder is less than $1/(12v)$.

Retain every integer r with $|r/k - X| \leq \sqrt{X}/2$. Their number is at least $(k - 1/3)\sqrt{X}$, because $X \geq 10$. On this block, $L \geq 5X/6$ and $\sqrt{L-1}/X \geq 5/(6\sqrt{X})$. Taylor's theorem, applied to $X - L + L\log(L/X)$ with second derivative $1/L$, gives

$$(k+1)[X - L + L\log(L/X)] \leq \frac{(k+1)(L-X)^2}{2\min(X,L)} \leq \frac{21}{20}, \qquad \frac{k+1}{12kL} \leq \frac{1}{80}.$$

Put $c_k = \sqrt{(k+1)/(2\pi k^2)}$. The rational estimates

$$e^{17/16} < 3, \qquad (k - 1/3)c_k > 4/5 \quad (k = 4, 5, 6)$$

follow from the exponential Taylor series with its geometric tail and $\pi < 22/7$. Dividing (5) by $Xe^{(k+1)X}\sqrt{k/(2\pi(k+1))}$ and summing therefore proves

$$\frac{Cm^{-3/2}W_+}{M} > \frac{2}{9}. \tag{6}$$

For $-1/2 \leq v < 0$, the identity $1/\Gamma(v) = v/\Gamma(v+1)$ and $\Gamma(v+1) \geq 1$ give $|1/\Gamma(v)| \leq 1/2$. The latter gamma inequality follows from convexity of $\log\Gamma$ and its equal values zero at 1 and 2. There are at most three nonzero terms in S, with $r = 0, 1, 2$. Since $z \leq 6X^{5/4}$,

$$S \leq (1 + z + z^2/2)/2 \leq 13X^{5/2}.$$

Equation (6), together with $\sqrt{k/(2\pi(k+1))} > 1/4$, implies $W_+ > \sqrt{X}e^{5X}/40$. Consequently

$$0 \leq S/W_+ < 520X^2e^{-5X} < 1/100 \qquad (X \geq 10).$$

The last expression decreases on this half-line, and its value at 10 is bounded using the positive exponential series. Combining with (6) proves $H/M > (99/100)(2/9) > 1/5$.

To obtain (4), retain the larger block $|L - X| \le \sqrt{X}$, and write $v = (L - X)/\sqrt{X}$. For $X \ge 15$, we have $1/\sqrt{X} < 4/15$ and $1/X \le 1/15$; thus $L \ge 11X/15 \ge 11$ and $\sqrt{(L-1)/X} > 4/5$. The Stirling correction is less than $1/100$, while the exponent loss is

$$(k+1)[X - L + L\log(L/X)] \le \frac{15(k+1)}{22}v^2 \le \frac{105}{22}v^2.$$

The values of v form a translated mesh of spacing $h = 1/(k\sqrt{X}) < 1/15$. For $f(v) = e^{-105v^2/22}\mathbf{1}_{[-1,1]}(v)$, every nonempty superlevel set is an interval. The number of mesh points in any interval of length ℓ is at least $\ell/h - 1$. Integrating this count over levels $0 < y < 1$ proves

$$h \sum_{v \text{ on mesh}} f(v) \ge \int_{-1}^{1} e^{-105v^2/22}\,\mathrm{d}v - h.$$

For $a = 105/22$, the elementary Gaussian tail bound gives

$$\int_{-1}^{1} e^{-av^2}\,\mathrm{d}v \ge \sqrt{\pi/a} - e^{-a}/a > \frac{81}{100} - \frac{1}{100} = \frac{4}{5}.$$

Here $\pi > 157/50$ proves the square-root inequality, and the positive Taylor series of e^a proves $e^a > 100/a$. Since $e^{-1/100} > 99/100$ and $kc_k = \sqrt{(k+1)/(2\pi)} > 89/100$, summing the bounds from (5) yields

$$\frac{H_+}{M} > \frac{4}{5}\frac{99}{100}\frac{89}{100}\left(\frac{4}{5} - \frac{1}{15}\right) = \frac{64614}{125000} > \frac{1}{2}.$$

This proves the stronger normalization with no limiting or sampled argument.

3.2 A variance argument

Normalize the positive summands as a probability distribution w_r on integers $r \ge r_0 = \lfloor k/2 \rfloor + 1$, and write $\mu = \mathbb{E}R$. The sequence

$$b_r = \frac{(r+1)w_{r+1}}{w_r} = z\frac{\Gamma(r/k - 1/2)}{\Gamma((r+1)/k - 1/2)}$$

decreases by gamma log-convexity. Thus $\mathrm{Cov}(R, b_R) \le 0$. Keeping the left boundary terms in the two expectations gives

$$\mathbb{E}b_R = \mu - r_0 w_{r_0}, \qquad \mathbb{E}(Rb_R) = \mathbb{E}[R(R-1)] - r_0(r_0 - 1)w_{r_0}.$$

It follows that

$$\mathbb{E}[R(R-1)] \le \mu^2 - r_0(\mu - r_0 + 1)w_{r_0} \le \mu^2, \qquad \mathrm{Var}(R) \le \mu.$$

This calculation does not assume a variance bound for a shifted distribution without checking its boundary.

For a gamma-distributed random variable Y of shape $u > 0$ and unit scale, concavity of $Y^{1/k}$ gives

$$\frac{\Gamma(u + 1/k)}{\Gamma(u)} = \mathbb{E}Y^{1/k} \le u^{1/k}.$$

Since $u \mapsto u^{-1/k}$ is convex, a second Jensen inequality yields

$$\mu \geq \mathbb{E}b_R \geq z(\mu/k - 1/2)^{-1/k}.$$

In particular $\mu^{k+1} \geq kz^k$ and $\mu \geq kX$. With $H_+ = Cm^{-3/2}W_+$, termwise differentiation now gives

$$-m^2(\log H_+)'' = \frac{\mu}{k} - \frac{3}{2} - \frac{\mathrm{Var}(R)}{k^2} \geq \frac{k-1}{k}X - \frac{3}{2} > 0. \tag{7}$$

Finally put $\epsilon = S/W_+$. Each summand of ϵ is a positive constant times $m^{r/k-3/2}/H_+$, where the exponent is negative. Its logarithm is convex by (7); hence the summand itself, and their sum ϵ, are convex. Since $0 \leq \epsilon < 1$,

$$(\log(1-\epsilon))'' = -\frac{\epsilon''}{1-\epsilon} - \frac{(\epsilon')^2}{(1-\epsilon)^2} \leq 0.$$

The identity $H = H_+(1-\epsilon)$ shows that subtracting the finitely many negative gamma terms can only strengthen the logarithmic curvature. This completes the proof of Theorem 3.1.

4 An explicit transformation and a local curvature test

In this section $k \in \{4, 5, 6\}$ and

$$A = \Gamma(1+1/k)\zeta(1+1/k), \quad C = (2\pi)^{-k/2}, \quad K = (2\pi)^{1+1/k}, \quad c = \zeta(-k)/2.$$

Thus $c = 0$ for $k = 4, 6$ and $c = -1/504$ for $k = 5$.

4.1 A transformation with an explicit remainder

Lemma 4.1. *For* $\mathrm{Re}\, z > 0$*, with principal powers and logarithms,*

$$\log F_k(e^{-z}) = Az^{-1/k} + \tfrac{1}{2}\log z - \tfrac{k}{2}\log(2\pi) + cz + R(z), \tag{8}$$

$$R(z) = \sum_{l,m\geq 1} \frac{1}{l} \sum_{h=0}^{k-1} \exp\left\{-Ke^{i(k-1-2h)\pi/(2k)} l(m/z)^{1/k}\right\}. \tag{9}$$

The last series converges absolutely. If

$$b(z) = K|z|^{-1/k}\cos\left(\frac{(k-1)\pi}{2k} + \frac{|\arg z|}{k}\right), \qquad E_k(b) = \frac{ke^{-b}}{1-e^{-b}} \sum_{j=0}^{k} \frac{k!}{(k-j)!b^j},$$

then $b(z) > 0$ *and* $|R(z)| \leq E_k(b(z))$.

Proof. Mellin inversion on a line $\mathrm{Re}\, s > 1/k$ gives the integrand $\Gamma(s)\zeta(1+s)\zeta(ks)z^{-s}$. Shift the line to $\mathrm{Re}\, s = -3/2$. The crossed poles at $1/k, 0, -1$ have residues $Az^{-1/k}$, $\frac{1}{2}\log z - \frac{k}{2}\log(2\pi)$ and cz, respectively; the last residue vanishes for even k. The horizontal integrals vanish by gamma decay and polynomial bounds for zeta on fixed vertical strips, since $|\arg z| < \pi/2$.

On putting $w = -ks$, the two zeta functional equations and gamma reflection give the meromorphic identity

$$\frac{1}{k}\Gamma(-w/k)\zeta(1-w/k)\zeta(-w) = \Gamma(w)\zeta(1+w)\zeta(w/k)K^{-w}\frac{\sin(\pi w/2)}{\sin(\pi w/(2k))}.$$

The sine quotient equals $\sum_{h=0}^{k-1}\exp(i(k-1-2h)\pi w/(2k))$. On $\operatorname{Re} w = 3k/2$, both Dirichlet series converge absolutely. Every argument $Ke^{i(k-1-2h)\pi/(2k)}z^{-1/k}$ has positive real part. Gamma decay therefore permits termwise Mellin inversion and gives (9), including the orientation change and factor $1/k$.

The least real part of those arguments is $b(z)$. Discarding $1/l \le 1$ and summing in l bounds $|R|$ by $k(1-e^{-b})^{-1}\sum_{m\ge1} e^{-bm^{1/k}}$. The decreasing summand satisfies

$$\sum_{m\ge1} e^{-bm^{1/k}} \le e^{-b} + \int_1^\infty e^{-bu^{1/k}}\,\mathrm{d}u = e^{-b}\sum_{j=0}^{k}\frac{k!}{(k-j)!b^j}.$$

This proves both the stated bound and absolute convergence. □

Put $m = n + c$, $t = (A/(km))^{k/(k+1)}$, $x = t^{-1/k}$ and $X = mt = Ax/k$. The model of Section 3 also has the Hankel representation

$$H(u) = \frac{C}{2\pi i}\int_{\mathcal{H}} \sqrt{z}\exp(Az^{-1/k} + uz)\,\mathrm{d}z, \qquad u > 0. \tag{10}$$

Here $\mathcal{H}$ goes from negative infinity below the negative real axis, encircles zero counterclockwise while staying a positive distance from it, and returns above the axis. Expanding the exponential on this contour and using the reciprocal-gamma Hankel integral proves (10): the terms are exactly $Cu^{r/k-3/2}A^r/(r!\Gamma(r/k-1/2))$. The expansion is uniformly dominated because the contour stays away from zero, and its infinite ends have exponential decay.

4.2 Certified elementary constants

For $r \ge 0$ write $\delta = 1/k$, $\beta = (k-1)\pi/(2k)$ and

$$B(r) = K\operatorname{Re}\{e^{-i\beta}(1+ir)^{-\delta}\}, \qquad D(r) = A[1 - \operatorname{Re}(1+ir)^{-\delta}].$$

The following bounds will be used:

$$\frac{101k}{100} < A < \frac{21k}{20}, \quad B(0) > 2, \quad B(3) > \frac{3}{8}, \quad \inf_{r\ge0}(D(r)+B(r)) > \frac{7}{5}, \quad D(3) > \frac{7}{6}, \tag{11}$$

and

$$\frac{A}{k} + A\left(1 - 3^{-1/k}\cos\frac{\pi}{2k}\right) > 2. \tag{12}$$

The real-axis bound is elementary: $K > 6^{7/6} > 8$, since $\pi > 3$ and $6^7 > 8^6$, while $\cos\beta = \sin(\pi/(2k)) \ge \sin(\pi/12) = (\sqrt{6}-\sqrt{2})/4 > 1/4$. Thus $B(0) > 2$ without special-function evaluation. These involve only three finite evaluations. In particular they do not depend on a numerical grid covering an infinite domain. To see this, put $a = A - K\cos\beta > 0$, $b = K\sin\beta > 0$ and $\theta_0 = (1+\delta)^{-1}\arctan(b/a)$. Direct differentiation, with $\theta = \arctan r$, gives

$$\inf_{r\ge0}(D(r)+B(r)) = A - \sqrt{a^2+b^2}\cos(\theta_0)^{1+\delta}. \tag{13}$$

Indeed, the subtracted function is $\cos^\delta\theta[a\cos(\delta\theta) + b\sin(\delta\theta)]$; its derivative has the sign of $b\cos((1+\delta)\theta) - a\sin((1+\delta)\theta)$, which changes once from positive to negative. Also B decreases on $[0,3]$, because its derivative is a negative multiple of $\sin(\beta + (1+\delta)\arctan r)$, positive on this interval.

For transparency, the following are outward rational enclosures of the four quantities on the left of the last three inequalities and (12). Each displayed interval has rational decimal endpoints.

k	$B(3)$	$\inf(D+B)$	$D(3)$	negative-ray loss
4	$(.59939, .59941)$	$(1.79026, 1.79029)$	$(1.19272, 1.19275)$	$(2.28243, 2.28246)$
5	$(.46350, .46353)$	$(1.58462, 1.58465)$	$(1.18249, 1.18252)$	$(2.24122, 2.24126)$
6	$(.37759, .37762)$	$(1.40266, 1.40270)$	$(1.17621, 1.17625)$	$(2.21504, 2.21509)$

The certificate evaluates $\Gamma(1+1/k)$ and $\zeta(1+1/k)$ and the closed expressions above with 256-bit Arb balls; rational comparisons use the whole enclosing ball, not its midpoint. It also verifies $A \in (4.16502, 4.16503)$, $(5.13401, 5.13403)$, $(6.11293, 6.11296)$, respectively. The executable verifier records the library version and full enclosures. Thus these are certified numerical inequalities, with Arb's rigorous real-ball arithmetic an explicit computational dependency.

For $x \geq 15$ and $|r| \leq 3$, $b(t(1+ir)) \geq 3x/8 \geq 45/8$. The formula of Lemma 4.1 gives, by replacing k by 6 and b by $45/8$,

$$E_k(b) < 23/250, \qquad \frac{6}{1-e^{-45/8}} \left(\sum_{j=0}^{6} \frac{6!}{(6-j)!(45/8)^j} \right) e^{E_6(45/8)} < 28.$$

The same ball certificate encloses these two left sides in $(.09119, .09121)$ and $(27.7010, 27.7012)$; they can alternatively be bounded with rational exponential Taylor series. Since $|e^R - 1| \leq |R| e^{|R|}$, we obtain

$$\left| C\sqrt{z}\, e^{Az^{-1/k}} (e^{R(z)} - 1) \right| \leq 28C\sqrt{t}(1+r^2)^{1/4} e^{Ax - 7x/5}. \tag{14}$$

Here the decay of the model and the remainder was combined before taking a supremum: its exponent is $-x[D(r) + B(r)]$.

4.3 First and second derivatives of the model

The notation H_+ and the probability weights from Section 3 are retained.

Lemma 4.2. *For $X \geq 10$, the mean of those weights satisfies $kX \leq \mu \leq 2kX + 3$, and $0 < H'_+(m)/H_+(m) \leq 2t$. Moreover*

$$-2t^2/X \leq (\log H_+)'' \leq 0.$$

Proof. The lower mean bound and the nonpositive logarithmic curvature were proved in (7). For $y > 0$ and $\delta = 1/k$, Jensen's inequality for a gamma variable of shape $y+\delta$ and the concave exponent $1-\delta$ gives

$$\Gamma(y+\delta)/\Gamma(y) \geq y/(y+\delta)^{1-\delta}.$$

Writing $s = kX$, the consecutive probability ratio therefore satisfies

$$\frac{w_{r+1}}{w_r} \leq \frac{s^{1+1/k}}{(r-k/2)(r+1)^{1/k}} \leq \frac{40}{39} 2^{-1-1/k} < \frac{1}{2} \qquad (r \geq 2s).$$

The last comparison follows by raising $40/39 < 2^{1/6}$ to the sixth power. The tail beyond $R_0 = \lceil 2s \rceil$ is geometrically dominated, so $\mathbb{E}(R - R_0)_+ \leq \sum_{j \geq 1} j 2^{-j} = 2$. Thus $\mu \leq R_0 + 2 \leq 2s + 3$. Termwise differentiation gives $mH'_+/H_+ = \mu/k - 3/2 \in (0, 2X]$. Finally $(\log H_+)'' = (\mathrm{Var}(R)/k^2 - \mu/k + 3/2)/m^2 \geq -2t^2/X$. □

Fix from now on $m \geq 100$, $x \geq 15$, and the associated t, X. Every auxiliary function of u below uses this same fixed t for $|u - m| \leq 1$. Put $h(u) = e^{-tu} H_+(u)$. The model parameter at u is $X_u = X(u/m)^{1/(k+1)}$, and $t_u/t = (m/u)^{k/(k+1)}$. Lemma 4.2 shows, on the entire interval,

$$|h'/h| \leq 2t, \qquad |h''/h| \leq 4t^2, \qquad -(\log h)'' \geq \frac{3t^2}{5X}, \qquad h(u) \geq \frac{100}{101} h(m). \tag{15}$$

Here is an explicit check of the uniform constants. Since $99/100 \le u/m \le 101/100$, we have $t_u/t \le 100/99$ and $X_u \ge (99/100)X$. The first bound follows from $0 < H_+'(u)/H_+(u) \le 2t_u$. The identity $h''/h = (h'/h)^2 + (\log H_+)''$, together with $-2t_u^2/X_u \le (\log H_+)'' \le 0$, proves the second. For the third, use (7), $X \ge x \ge 15$, and

$$\frac{X_u/u^2}{X/m^2} \ge \frac{99/100}{(101/100)^2}, \qquad \frac{k-1}{k} - \frac{3}{2X_u} \ge \frac{3}{4} - \frac{100}{990}.$$

Their product is greater than $3/5$. Finally integrate $|h'/h| \le 2t$ and use $t \le 15^{-4}$ and $e^{2t} \le (1-2t)^{-1} < 101/100$ to get $h(u) \ge e^{-2t}h(m) > 100h(m)/101$. The weaker comparison $h(u) > h(m)/2$ is sometimes retained below to simplify already small error constants.

4.4 Explicit errors with the local parameter fixed

Let $J(u)$ be the inverse-Laplace integral on $t-3it$ to $t+3it$ of $C\sqrt{z}e^{Az^{-1/k}}(e^{R(z)}-1)e^{uz}$. Tilting by e^{-tu} makes its jth derivative acquire $(z-t)^j$. From (14), the strengthened bound $H_+(m) > M(m)/2$, $\sqrt{2(k+1)/(\pi k)} < 1$, and $(1+r^2)^{1/4} < 9/5$, it follows that

$$\frac{|(e^{-tu}J(u))^{(j)}|}{t^j h(u)} \le \frac{1008}{5}\sqrt{X}\,\frac{3^{j+1}}{j+1}e^{-7x/5}, \qquad j = 0,1,2, \quad |u-m| \le 1. \tag{16}$$

At $u = m$ the constant is $504/5$; the factor 2 accounts for the last bound in (15). The absolute value of the tilted vertical integrand is independent of u.

Let $V(u)$ be the corresponding vertical model integral without $e^R - 1$. Deform the Hankel contour to this segment and the two horizontal rays $z = t(v \pm 3i)$, $v \le 1$. No deformation crosses zero or the negative real cut, and the connections at negative infinity vanish. For $0 \le v \le 1$ the real model phase has derivative

$$\frac{\mathrm{d}}{\mathrm{d}v}\operatorname{Re}\{v + k(v+3i)^{-1/k}\} = 1 - \operatorname{Re}(v+3i)^{-1-1/k} > 2/3.$$

Its endpoint loss is $D(3)x > (7/6)x$. Also $|v+3i|^{1/2} < 9/5$ and $|v-1+3i| < 16/5$. For $v = -w \le 0$, the function $\operatorname{Re}(v+3i)^{-1/k}$ increases in v: its derivative is $-\operatorname{Re}(v+3i)^{-1-1/k}/k > 0$. Thus the negative-ray loss is at least $2x + Xw$, by (12), while $|v+3i|^{1/2} \le w+2$ and $|v-1+3i| \le w+4$. Write

$$P_j(w) = (w+2)(w+4)^j = \sum_l p_{j,l}w^l, \qquad I_j(X) = \int_0^\infty P_j(w)e^{-Xw}\,\mathrm{d}w = \sum_l \frac{p_{j,l}l!}{X^{l+1}}.$$

Integration and the same model lower bound give, uniformly on the local interval,

$$\frac{|(e^{-tu}(H(u)-V(u)))^{(j)}|}{t^j h(u)} \le \frac{108}{5}(16/5)^j X^{-1/2}e^{-7x/6} + 16\sqrt{X}e^{-2x}I_j(X), \qquad j = 0,1,2. \tag{17}$$

For clarity, the constants before extending from the center to the interval are $27/5$ and 2. On the positive ray, replacing m by $u \ge m-1$ leaves phase derivative greater than $1/2$, so the integral factor $3/(2X)$ is replaced by at most $2/X$; the denominator changes by at most 2. This is covered by the factor 4. On the negative ray, replace the moment parameter X by $X - t \ge (99/100)X$ and multiply the endpoint exponential by $e^t < 2$. As P_j has degree at most 3, $I_j(X-t) \le (100/99)^4 I_j(X) < 2I_j(X)$. Including the denominator factor 2 gives the displayed factor 8.

The finite negative part of the model is also explicitly controlled. Put $S_H(u) = H_+(u) - H(u)$; its powers have exponents $-3/2 \le a < -1$. The bound in Section 3 and $X_u \ge (99/100)X$ give

$$S_H(u)/H_+(u) \le 520X^2 e^{-99X/20}.$$

For each power u^a, the first tilted derivative has absolute factor $t + |a|/u < 2t$, and the second has factor $t^2 - 2ta/u + a(a-1)/u^2 < 2t^2$, since $ut \ge (99/100)X \ge 14$. Consequently, for $j = 0, 1, 2$,

$$\frac{|(e^{-tu}S_H(u))^{(j)}|}{t^j h(u)} \le 1040X^2 e^{-99X/20}. \tag{18}$$

4.5 The precise remaining arc criterion

For fixed integer n, let $m = n + c$ and use the fixed t above to define the real coefficient interpolation

$$Q(u) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F_k(e^{-t-i\theta})e^{(u-c)(t+i\theta)}\,\mathrm{d}\theta. \tag{19}$$

Conjugation makes this real for real u, and Cauchy's formula gives $Q(m+j) = p_k(n+j)$ for every integer j with $n + j \ge 0$. Let $L(u)$ denote the portion with $3t \le |\theta| \le \pi$. The exact transformation and Hankel deformation give

$$Q = H_+ - S_H + J - (H - V) + L.$$

Thus every error term except L has just been bounded explicitly.

Proposition 4.3. *Suppose bounds $\varepsilon_j \ge 0$, $j = 0, 1, 2$, hold uniformly for $|u - m| \le 1$ in the form*

$$|(e^{-tu}Q(u) - h(u))^{(j)}| \le \varepsilon_j t^j h(u).$$

If $\varepsilon_0 < 1$ and

$$4\varepsilon_1 + \varepsilon_2 + 4\varepsilon_0 + \varepsilon_0\varepsilon_2 < \frac{3}{5X}, \tag{20}$$

then $D_k(n) > 0$.

Proof. Write $e = e^{-tu}Q - h$. Positivity follows from $|e| < h$. Expand $-(h+e)(h''+e'') + (h'+e')^2$ and discard the nonnegative e'^2. Using (15), the result is at least

$$h^2t^2\left(\frac{3}{5X} - 4\varepsilon_1 - \varepsilon_2 - 4\varepsilon_0 - \varepsilon_0\varepsilon_2\right) > 0.$$

Therefore $\log Q$ has strictly negative second derivative on this interval. The exact triangular-kernel identity

$$2f(m) - f(m-1) - f(m+1) = -\int_{-1}^{1}(1 - |v|)f''(m+v)\,\mathrm{d}v$$

for $f = \log Q$ and the interpolation values prove the assertion. □

The remaining quantities have an exact Euler-product expression. Put

$$\rho(t,\theta) = \frac{|F_k(e^{-t-i\theta})|}{F_k(e^{-t})} = \prod_{l\ge 1}\left(1 + \frac{\sin^2(l^k\theta/2)}{\sinh^2(l^kt/2)}\right)^{-1/2}, \qquad J_j^{\mathrm{out}}(t) = \int_3^{\pi/t} r^j\rho(t,tr)\,\mathrm{d}r.$$

Taking absolute values in (19) after tilting and differentiating, using symmetry, the transformation on the real axis and $H_+ > M/2$, gives the uniform local bound

$$\frac{|(e^{-tu}L(u))^{(j)}|}{t^j h(u)} \le \frac{101}{50}\sqrt{X}\, e^{R(t)} J_j^{\mathrm{out}}(t), \qquad j = 0, 1, 2. \tag{21}$$

The sharper factor 101/100 in the local model comparison is included. Equations (16), (17), (18) and (21) specify a fully explicit sufficient test. The outer moments must be proved on an unbounded range of x before Proposition 4.3 becomes an infinite-tail theorem; the proposition alone does not assert that they meet the test.

As a numerical scale for this last obligation, let a_j be the sum of the three right sides in (16), (17) and (18). For every $x \ge 15$,

$$X(4a_0 + 4a_1 + a_2 + a_0a_2) < 1/2000. \tag{22}$$

To verify this on the whole half-line, use $x \le X \le 21x/20$ in positive prefactors and negative powers separately. Each resulting xa_j decreases for $x \ge 15$, as does xa_0a_2; this follows by logarithmic differentiation of the finitely many terms $x^b e^{-cx}$, whose exponents have $b \le 5$ and $c \ge 7/6$. At $x = 15$, the resulting upper bounds for (a_0, a_1, a_2) are respectively $(.000001961, .000003179, .000006895)$, and the upper bound for the left side of (22) is $.000433 < 1/2000$. These comparisons, with the exact expressions and the moments $I_0 = 2/X + 1/X^2$, $I_1 = 8/X + 6/X^2 + 2/X^3$ and $I_2 = 32/X + 32/X^2 + 20/X^3 + 6/X^4$, are included in the ball certificate. Cross terms with the outer-arc errors must still be included when using (20).

5 Weighted bounds for the noncentral arcs

For $t > 0$, put $x = t^{-1/k}$ and

$$\rho_k(t, \theta) = \frac{|F_k(e^{-t+i\theta})|}{F_k(e^{-t})}.$$

The modulus of each Euler factor gives the exact identity

$$\rho_k(t, \theta) = \prod_{j \ge 1} \left(1 + \frac{\sin^2(j^k\theta/2)}{\sinh^2(j^k t/2)}\right)^{-1/2}. \tag{23}$$

Each factor is at most one, so retaining any finite subset gives an upper bound. We retain the first two angular moments as well as the unweighted integral, because these moments enter the curvature test.

5.1 Intermediate arcs

Set $\delta_k = 6^{-k}$. For $r = 0, 1, 2$, define

$$J_{k,r}(x) = \frac{1}{t}\int_{3t}^{\delta_k} (\theta/t)^r \rho_k(t, \theta)\, \mathrm{d}\theta.$$

Lemma 5.1. *For every real $x \ge x_0$, the bounds $x^2 J_{k,r}(x) \le C_{k,r}$ hold with the following constants. All displayed terminating decimals are exact rational numbers.*

k	x_0	$C_{k,0}$	$C_{k,1}$	$C_{k,2}$
4	18	0.00023126	0.00084142	0.00501233
5	15	0.00111779	0.00386236	0.01506805
6	15	0.00057447	0.00195712	0.00696895

Proof. For $0 \le u \le 1/2$, the sine series gives $\sin u/u \ge 23/24$. For $0 \le v \le 1/6$, successive terms in the nonconstant part of $\sinh v/v$ have ratio at most $1/720$, whence

$$\frac{\sinh v}{v} \le 1 + \frac{1/216}{1 - 1/720} < \frac{201}{200}, \qquad \frac{23/24}{201/200} > \frac{19}{20}.$$

Write $y = \theta/t \ge 3$ and $l = \lfloor \theta^{-1/k} \rfloor$. For $j \le l$, we have $j^k\theta \le 1$ and $j^k t \le 1/3$. Therefore

$$\frac{|\sin(j^k\theta/2)|}{\sinh(j^k t/2)} \ge \frac{19}{20}y, \qquad \rho_k(t,\theta) \le [1 + (19y/20)^2]^{-l/2} \le (20/(19y))^l.$$

Here $l \ge 6$. On the subinterval with this value of l, the variable y lies between $(x/(l+1))^k$ and $(x/l)^k$, up to endpoints of measure zero. Extending its upper endpoint to infinity yields

$$J_{k,r}(x) \le \sum_{l=6}^{\infty} \frac{(20/19)^l}{l - r - 1} \max\{3, (x/(l+1))^k\}^{r+1-l}. \tag{24}$$

After multiplication by x^2, each summand first increases as x^2 and then decreases as $x^{2-k(l-r-1)}$. Its maximum on $x \ge x_0$ is at x_0 if $3(l+1)^k \le x_0^k$, and otherwise at $x = 3^{1/k}(l+1)$. Since $3^{2/k} \le 2$, a rational upper bound is

$$B_l = \frac{(20/19)^l}{l - r - 1} \begin{cases} x_0^2((l+1)/x_0)^{k(l-r-1)}, & 3(l+1)^k \le x_0^k, \\ 2(l+1)^2 3^{r+1-l}, & 3(l+1)^k > x_0^k. \end{cases}$$

The second expression also bounds the maximum in the first case. Sum the displayed bounds for $6 \le l < 100$. For $l \ge 100$, the ratio of consecutive second expressions is at most

$$\frac{20}{57}\left(\frac{102}{101}\right)^2 < \frac{9}{25}.$$

Thus the remaining infinite sum is at most $25/16$ times its $l = 100$ term. Exact fraction arithmetic gives the table after rounding upward. These finite rational calculations are specified completely by the formulas above and implemented in `minor_intermediate.py`. □

5.2 Far arcs and their exact interval certificates

For the same pairs (k, x_0), put $t_0 = x_0^{-k}$ and

$$I_{k,r}(t) = \int_{\delta_k}^{\pi} \theta^r \rho_k(t,\theta)\, \mathrm{d}\theta.$$

Lemma 5.2. *For every real $0 < t \le t_0$ and $r = 0, 1, 2$,*

$$I_{k,r}(t) \le B_{k,r}(t/t_0)^{7/2}, \tag{25}$$

with the following rational upper bounds.

k	x_0	$B_{k,0}$	$B_{k,1}$	$B_{k,2}$	*cells*
4	18	$7.864705 \cdot 10^{-17}$	$3.574880 \cdot 10^{-18}$	$1.5 \cdot 10^{-18}$	7855
5	15	$2.111634 \cdot 10^{-18}$	$4.059085 \cdot 10^{-21}$	$5 \cdot 10^{-21}$	81633
6	15	$2.380075 \cdot 10^{-21}$	$9.299098 \cdot 10^{-25}$	$1.2 \cdot 10^{-24}$	456399

Proof. We first describe the entire finite certificate and its verification. Let $\alpha = \theta/(2\pi)$, $B = 44 \cdot 6^k$, and $W = B/2 - 7$. Because $\pi < 22/7$, the required α interval is contained in $[7/B, 1/2]$. A certificate leaf (l, i) represents the closed interval

$$\left[\frac{7 \cdot 2^l + iW}{B2^l}, \frac{7 \cdot 2^l + (i+1)W}{B2^l}\right]. \tag{26}$$

The leaves are checked for exact consecutive coverage by putting their dyadic coordinates $[i/2^l, (i+1)/2^l]$ over the common denominator $2^{\max l}$. They cover $[0, 1]$ with no gaps or overlapping interiors. Shared closed endpoints have measure zero.

For each cell and $1 \le j \le J := \lfloor 3x_0/2 \rfloor$, let d_j be the minimum distance of $j^k\alpha$ from an integer over the cell. If the multiplied closed interval contains an integer, $d_j = 0$; otherwise d_j is the smaller endpoint distance. This is an exact rational calculation, and $|\sin(\pi j^k \alpha)| \ge \sin(\pi d_j)$ with $0 \le d_j \le 1/2$.

The rational numbers

$$\pi_- = \frac{3141592653589793}{10^{15}}, \qquad \pi_+ = \frac{3141592653589794}{10^{15}}$$

bound π. This is checked from Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and alternating series through 30 and 10 terms. The identity follows from the tangent addition formula with the angles in the first quadrant. At the grid points $y = i/2048$, $0 \le i \le 1024$, a sine lower bound is

$$L(y) = \pi_- y - \frac{(\pi_+ y)^3}{3!} + \frac{(\pi_- y)^5}{5!} - \frac{(\pi_+ y)^7}{7!} + \frac{(\pi_- y)^9}{9!} - \frac{(\pi_+ y)^{11}}{11!}.$$

Round this downward to a multiple of 2^{-40}, taking the endpoint values 0 and 1 exactly. Linear interpolation of these lower endpoint values stays below $\sin(\pi y)$, by its concavity on $[0, 1/2]$. This gives a rational $s_j \le \sin(\pi d_j)$.

For $v = j^k/(2x_0^k)$, an upper bound for $\sinh v$ is

$$\sum_{a=0}^{20} \frac{v^{2a+1}}{(2a+1)!} + \frac{v^{43}/43!}{1 - v^2/(44 \cdot 45)}.$$

Indeed, the denominator is positive for the stated parameters and bounds the ratios after the first omitted term. Round upward to a multiple of 2^{-60}, obtaining $u_j \ge \sinh v$. Set $a_j = (s_j/u_j)^2$, $A_0 = 2^{20}$, and $b_j = \lfloor A_0 a_j \rfloor$. Then

$$\rho_k(t_0, \theta) \le \left(\prod_{j=1}^{J} (1 + b_j/A_0)\right)^{-1/2}.$$

Writing $P = \prod_j (A_0 + b_j)$, the saved rational bound $U = z/2^h$ is verified by the integer inequality

$$z^2 P \ge 2^{20J+2h}. \tag{27}$$

The generator selects a 16-bit upward mantissa with an integer square root. The independent checker imports no generator code, recalculates the constants, uses the stronger 24-bit factor rounding, and verifies the corresponding version of (27) by multiplication.

For a cell with upper endpoint α_+, its contribution to the rth moment is at most

$$\frac{44W}{7B2^l} U \left(\frac{44\alpha_+}{7}\right)^r, \tag{28}$$

since $d\theta = 2\pi\, d\alpha$ and $2\pi < 44/7$. Adding these exact fractions gives values below the table. The supplement contains every leaf and the exact sums. The checker verifies all leaves, full coverage, and the sums. Floating-point priorities are used only to choose the generator's refinement order; every bound and stopping decision is rational.

For uniformity in t, each leaf is verified to have at least four factors with $b_j/A_0 \geq 7$. Convexity and $\sinh 0 = 0$ imply

$$\sinh(j^k t/2) \leq (t/t_0)\sinh(j^k t_0/2) \qquad (0 < t \leq t_0).$$

For $a \geq 0$ and $V \geq 1$, integration in $\log V$ gives

$$\log(1 + Va) - \log(1 + a) \geq \frac{a}{1+a}\log V.$$

Apply this to the exact squared ratios in (23), with $V = (t_0/t)^2$. Four factors contribute at least $4(7/8) = 7/2$ to the exponent of t/t_0; the other factors contribute nonnegative exponents. Consequently $\rho_k(t,\theta) \leq U(t/t_0)^{7/2}$ on each cell. Integrating proves (25) on the entire unbounded range $x \geq x_0$. □

For later use, after normalization by $\sqrt{x}/t^{r+1}$, the ratio of a far-moment error to the curvature scale $1/x$ decreases with x. Its power of x is $k(r+1-7/2)+3/2 < 0$ for $r \leq 2$ and $k \geq 4$. The intermediate errors have the same property because $\sqrt{x}J_{k,r}(x) \leq C_{k,r}x^{-3/2}$.

6 The explicit infinite tails and the finite join

We now assemble the analytic estimates, including every cross term in the local curvature criterion. Set

$$(x_{0,4}, x_{0,5}, x_{0,6}) = (18, 15, 15), \qquad (B_4, B_5, B_6) = (2000000, 12000000, 180000000).$$

Theorem 6.1. *For $k = 4, 5, 6$, one has $D_k(n) > 0$ for every integer $n \geq B_k$.*

For use in the proof, the moment bounds established in Lemmas 5.1 and 5.2 are recorded with simple rational upper endpoints. With $x_0 = x_{0,k}$, $t_0 = x_0^{-k}$ and $\delta = 6^{-k}$, they say, for every $x \geq x_0$ and $t = x^{-k}$,

$$\int_3^{\delta/t} r^j \rho(t, tr)\,\mathrm{d}r \leq C_{k,j}/x^2, \tag{29}$$

$$\int_\delta^\pi \theta^j \rho(t,\theta)\,\mathrm{d}\theta \leq (t/t_0)^{7/2} I_{k,j}, \qquad j = 0, 1, 2. \tag{30}$$

The values needed are

k	x_0	$C_{k,0}$	$C_{k,1}$	$C_{k,2}$
4	18	.00023126	.00084142	.00501233
5	15	.00111779	.00386236	.01506805
6	15	.00057447	.00195712	.00696895

and

k	$I_{k,0}$	$I_{k,1}$	$I_{k,2}$
4	$7.864705 \cdot 10^{-17}$	$3.574880 \cdot 10^{-18}$	$1.5 \cdot 10^{-18}$
5	$2.111634 \cdot 10^{-18}$	$4.059085 \cdot 10^{-21}$	$5 \cdot 10^{-21}$
6	$2.380075 \cdot 10^{-21}$	$9.299098 \cdot 10^{-25}$	$1.2 \cdot 10^{-24}$

All terminating decimals in these tables denote exact rationals rounded upward from the certificate values. They are not floating-point estimates.

Proof of Theorem 6.1. First $e^{R(t)} < 101/100$ throughout the range. Indeed, $b(t) = B(0)x > 2x \geq 30$. Set

$$q = \left(\sum_{r=0}^{60} \frac{30^r}{r!} \right)^{-1}, \qquad E_* = \frac{6q}{1-q} \sum_{j=0}^{6} \frac{6!}{(6-j)!30^j}.$$

The positive exponential series and Lemma 4.1 give $|R(t)| \leq E_6(30) \leq E_* < 1/101$. The last inequality is a rational comparison. Hence $e^{R(t)} \leq e^{|R(t)|} \leq (1-E_*)^{-1} < 101/100$.

Let a_j be the central errors of (22), and let b_j be an upper bound for the remaining relative derivative error in (21). Splitting the outer integral at δ and using (29)–(30) yields

$$b_j \leq \frac{101}{50}\frac{101}{100}\sqrt{X}\left\{ \frac{C_{k,j}}{x^2} + t^{-j-1}(t/t_0)^{7/2} I_{k,j} \right\}. \tag{31}$$

Here $X = (A/k)x$, so the first contribution to Xb_j decreases as $x^{-1/2}$. The second decreases as

$$x^{3/2-k(5/2-j)}, \qquad j = 0, 1, 2;$$

all three exponents are strictly negative for $k = 4, 5, 6$. Thus this is an estimate on the whole half-line, with its maximum at $x = x_0$.

Put $R_4 = 87/20$ and $R_5 = R_6 = 4$, so that $R_k^2 > 21x_{0,k}/20$. Define the following explicit rational numbers:

$$\beta_{k,j} = \frac{101}{50}\frac{101}{100}\frac{21x_0}{20} R_k \left\{ \frac{C_{k,j}}{x_0^2} + x_0^{k(j+1)} I_{k,j} \right\}. \tag{32}$$

Since $X < 21x/20$, the preceding monotonicity proves $Xb_j \leq \beta_{k,j}$ for every $x \geq x_0$.

The central budget gives $Xa_0 < 1/8000$ and $Xa_2 < 1/2000$ as well as $X(4a_0 + 4a_1 + a_2 + a_0a_2) < 1/2000$. With $\varepsilon_j = a_j + b_j$, all cross terms are therefore bounded by

$$\begin{aligned} X(4\varepsilon_0 + 4\varepsilon_1 + \varepsilon_2 + \varepsilon_0\varepsilon_2) < \frac{1}{2000} + 4\beta_{k,0} + 4\beta_{k,1} + \beta_{k,2} + \frac{\beta_{k,0}\beta_{k,2}}{x_0} \\ + \frac{\beta_{k,2}}{8000x_0} + \frac{\beta_{k,0}}{2000x_0}. \end{aligned} \tag{33}$$

Substitution of the terminating rationals in the two tables gives the following outward rational bounds for the right side:

k	upper bound in (33)	required upper bound
4	.29640838	3/5
5	.30192167	3/5
6	.23822163	3/5

These comparisons use integer fractions only. Positivity is also explicit: $\varepsilon_0 \leq (1/8000 + \beta_{k,0})/x_0 < 1$. Thus Proposition 4.3 proves $D_k(n) > 0$ whenever $m = n + c \geq 100$ and $x \geq x_{0,k}$.

Finally, $m = (A/k)x^{k+1}$ and $A/k < 21/20$. The exact endpoint comparisons are

$$\begin{aligned} 2000000 &> (21/20)18^5 = 1984046.4, \\ 12000000 - 1/504 &> (21/20)15^6 = 11960156.25, \\ 180000000 &> (21/20)15^7 = 179402343.75. \end{aligned}$$

Consequently $n \geq B_k$ implies the required $m \geq 100$ and $x \geq x_0$. This proves all three strict infinite tails, including their first integer endpoints. □

The analytic theorem requires a finite check only through $B_k - 1$, using coefficients through B_k. The executed finite calculation in Proposition 2.1 goes one index farther: it checks determinants through B_k and uses coefficients through $B_k + 1$. Thus the two proved ranges overlap at B_k, with both neighbors of that determinant included. The exact last-failure witnesses and equality lists from that finite calculation therefore combine with this strict tail without any untested integer.

The arithmetic verifier regenerates the rational intermediate sums, reads the exact rational far moments, checks their upward rounding to the displayed table, retains every term in (33), and verifies the three endpoint comparisons. Its report records the hash of each far certificate. The cellwise validity and coverage of those certificates are independently checked; neither a hash nor a successful arithmetic assembly is a substitute for that verification.

Proof of Theorem 1.1. Proposition 2.1 proves $D_k(n) > 0$ for every integer $T_k \le n \le B_k$, with the displayed values of T_k and B_k. Theorem 6.1 proves the same inequality for every $n \ge B_k$. These ranges overlap at B_k, so their union includes every integer from the proposed threshold onward. The finite computation uses coefficients through $B_k + 1$, and hence includes both neighbors of that overlap determinant. The displayed negative value at $T_k - 1$ excludes every smaller non-strict or strict threshold. Finally all equality indices in the finite range lie below T_k, and the strict infinite tail excludes any further equality. This proves the asserted thresholds and the global equality classification. □

7 Reproducibility and contribution

The source supplement contains the exact counting engine, reverse-order full reconstructions, a logarithmic-derivative prefix checker, compact signed determinant streams, and analytic checks using rational arithmetic and certified real intervals. The mathematical meanings of these computations are stated in the preceding proofs; a filename is not used as a substitute for an argument. Certificate hashes identify artifacts and do not prove a counting identity or determinant sign. The entry point `src/verify_all.py` checks the input manifest, reconstructs all three finite coefficient ranges, compares every determinant sign, validates the complete far-arc partitions, and recomputes the analytic joins. Its explicitly named analytic-only mode is a partial check.

The contribution beyond the cited eventual theorem is the explicit, computationally certified passage to these three sharp thresholds. The curvature model, transformation, interpolation and arc estimates are unified in k; their numerical certificates and finite ranges are specific to $k = 4, 5, 6$. We claim neither thresholds for other powers nor a priority guarantee beyond the recorded literature search. The proof depends on exact program execution and certified interval arithmetic, as a computer-assisted proof does; it is not a proof-assistant derivation of those implementations.

GPT-6 Astra, an AI system, developed the arguments, code, and manuscript and executed the computations. Separate internal reviews audited portions of the work. This is internal review, not independent human peer review. No proof-assistant formalization is claimed.

References

[1] A. Roy, *Log-Concavity and the Multiplicative Properties of Restricted Partition Functions*, Annals of Combinatorics, published 29 September 2026. https://doi.org/10.1007/s00026-026-00847-5.

[2] A. Roy, same title, arXiv:2404.03153v4, 10 May 2025. https://arxiv.org/abs/2404.03153v4.

[3] C. O'Sullivan, *Detailed asymptotic expansions for partitions into powers*, International Journal of Number Theory **19**(9) (2023), 2163–2196. https://arxiv.org/abs/2205.13468v2.